

PROJECT AND TEAM INFORMATION

Multi-Drone Search & Rescue for Dynamic Disaster Response Using Graph - Based Path Planning and Priority Task Allocation.

A smart DA and OOP based system for coordinating multiple drones during search and rescue operation through efficient mission allocation and route planning

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

In an emergency, search teams may have to cover a large area where some places are unsafe or difficult to reach. Drones can quickly check different areas from the air, but handling many drones at the same time can become difficult. Our project will create a simple software system that helps decide which drone should search which area and which route it should follow. It will consider mission priority, drone position, battery level, availability and basic drone capability. The main aim is to make the search process faster, more organized and easier to manage.

State of the Art / Current Solution

Today, drones are already used for aerial observation, mapping and emergency response, while operators use ground-control and mission-planning software to create routes and monitor vehicles. These systems can work with waypoints and simulated vehicles. Our PBL does not try to build a real drone or replace professional rescue software. Instead, we will build the decision-making part as a student-level C/C++ simulation. This lets us test the main DSA ideas safely on a normal computer before considering any advanced drone simulator.

Project Goals and Milestones

Our goal is to build a working prototype that can manage several simulated drones and rescue missions.

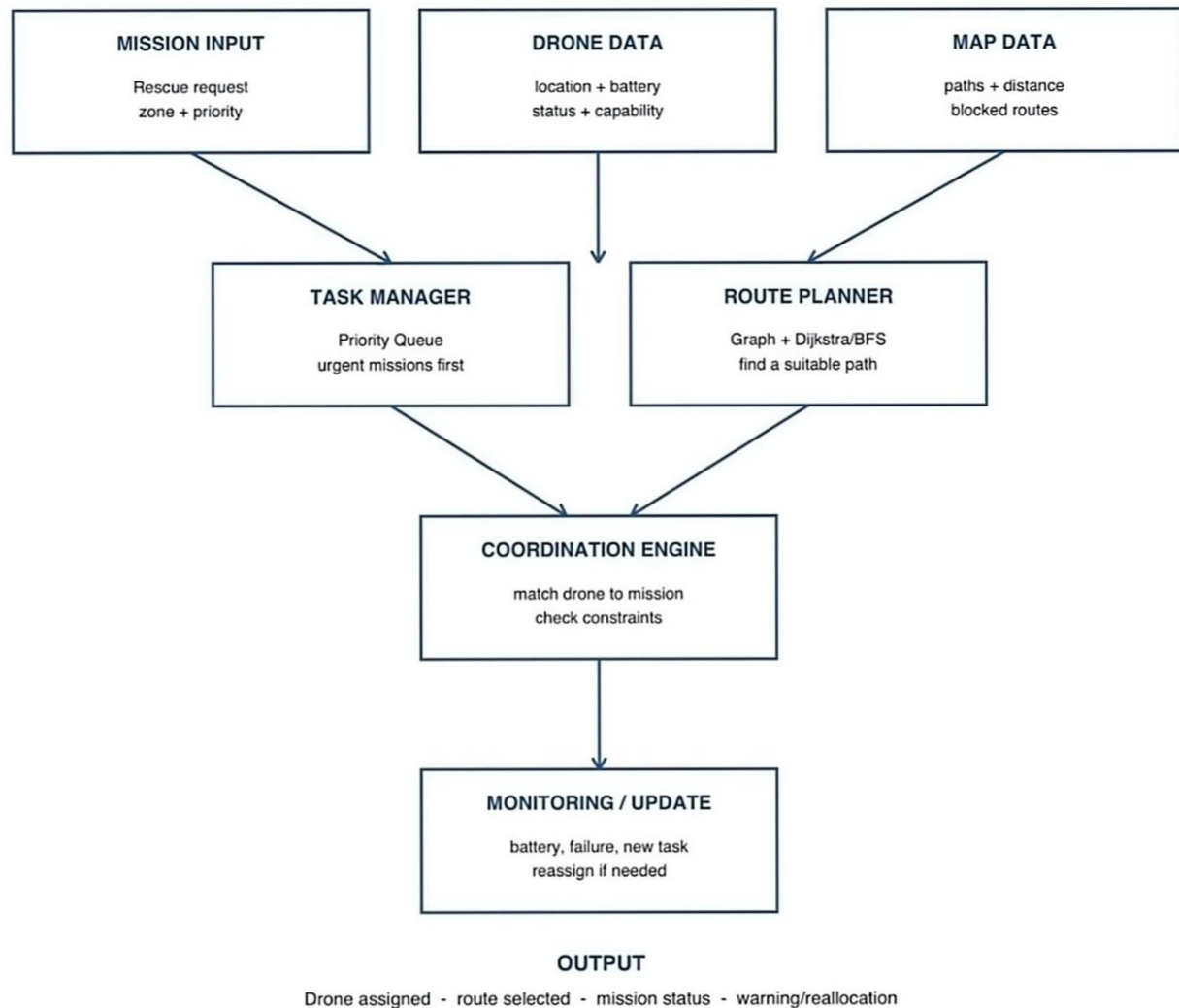
- Create data for drones, search zones, missions and map paths.
- Represent the search area using a graph.
- Use a priority queue to handle urgent missions first.
- Use BFS/Dijkstra to find a suitable route.
- Select a drone using simple factors such as distance, battery and availability.
- Reassign a mission if a drone becomes unavailable, its battery becomes low, a path is blocked, or a new urgent mission appears.
- Test the system with small emergency scenarios and show the final assignment and route.

Project Approach

The project will first be made as a C/C++ program with a simple menu or console interface. The map will be stored as a graph, where locations are nodes and possible flight paths are edges. Each path can have a distance or cost. Dijkstra's algorithm can then find a low-cost route between two locations. A priority queue will store rescue missions according to urgency, so a high-priority mission is handled before a normal one. C++ classes will represent objects such as Drone, Mission, SearchZone, Route and Coordinator. The Coordinator will check the available drones and assign one to a suitable mission. During testing, we can change battery levels, block a route, add a new emergency or mark a drone unavailable and then run the allocation again. If the basic version works well, the same idea can later be tested with a drone simulator such as ArduPilot SITL.

SYSTEM ARCHITECTURE (HIGH LEVEL DIAGRAM)

How the system works



Project Outcome / Deliverables

- A working C/C++ prototype for coordinating simulated drones.
- Graph-based map and route calculation.
- Priority-based rescue mission handling.
- Drone selection using simple practical conditions.
- Reallocation when a drone fails or a mission changes.
- Test results showing which drone was assigned and which route was selected.

Assumptions

- Drones are simulated; no physical drone is required for the first version.
- The search area is represented as a small weighted graph.
- Battery is represented by a simple percentage/value.
- Drone capabilities and mission requirements are predefined.
- Distances are simplified for a student-level simulation.
- The project focuses on planning and coordination, not real-time flight control.

References

1. ArduPilot Documentation - Simulation / SITL and Mission Planning: basic reference for software-based drone simulation and waypoint missions.
2. GeeksforGeeks - Dijkstra's Algorithm: simple reference for shortest-path calculation in a weighted graph.
3. GeeksforGeeks - Priority Queue in C++ STL: reference for priority-based mission handling.
4. cppreference - Classes: reference for designing the C++ classes used in the project. These are mainly learning and implementation references. The project will be built at a student level and will not depend on advanced AI or expensive drone hardware.